Scaling the Generative Quantum Eigensolver to 40 Qubits: Phase 3 Applied Execution and Validation

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1 Focus Area and Recap of Phase 2 Proposal

This Phase 3 submission executes and validates the Generative Quantum Eigensolver (GQE) pipeline proposed in Phase 2 for predicting the singlet–triplet gap ΔE_{ST} of 4CzIPN-class thermally activated delayed fluorescence (TADF) emitters, a materials class central to Mitsubishi Chemical Group's next-generation OLED programs. The target active space, CAS(20,20) under Jordan–Wigner (JW) mapping, requires exactly 40 qubits – beyond the reach of full state-vector simulation (2^{40} amplitudes, 16 TB) and the $\sim$10–12 qubit ceiling of existing GQE demonstrations. The proposed approach combines (i) a symmetry-preserving Givens-rotation ansatz that conserves particle number $\hat{N}$ and spin projection $\hat{S}_z$ by construction, (ii) Matrix Product State (MPS) energy evaluation via NVIDIA CUDA-Q's `tensornet-mps` backend (or an exact statevector fallback for validation-scale systems), and (iii) a Transformer generative model $\mathscr{G}_\phi$ that emits circuit parameters θ from the molecular Hamiltonian, trained end-to-end via energy-gradient backpropagation. A size-progressive transfer-learning curriculum (H_2 $\rightarrow$LiH$\rightarrow$$BeH_2$ $\rightarrow$$H_2O$$\rightarrow$4CzIPN) was proposed to avoid training the 40-qubit model from scratch. Full method and architecture detail is in our Phase 2 proposal; this write-up reports what was actually executed and validated in Phase 3, on qBraid, against that plan.

2 What Was Executed in Phase 3

All results below were produced by running `GQE_MPS_Transformer_final.ipynb` on a qBraid instance (CPU-only session; no GPU was allocated for this run, so all systems fell back to the exact `qpp-cpu` statevector backend rather than `tensornet-mps`). We report results exactly as obtained, including where the pipeline did not reach its target, per the challenge's guidance to be specific about numbers and honest about limitations.

2.1 Physics Correctness Validation

Before any learned component, we verified that a single particle-conserving Givens rotation reaches the exact ground state of the textbook 2-qubit reduced H_2/STO-3G Hamiltonian (O'Malley et al., 2016): FCI energy -1.762699 Ha vs. Givens-ansatz minimum -1.762698 Ha, an error of **0.0007 mHa**. This confirms the ansatz formulation and energy-evaluation code path are correct independent of the learned generator, isolating any downstream training issues to the optimization loop rather than the physics.

2.2 QCI Dirac-3 Platform Connectivity

As a secondary platform integration for future polynomial-optimization sub-problems, we instantiated the QCI `qci-client` against the Dirac-3 cloud endpoint and confirmed successful authentication with a live allocation balance of **259,200 s**. This validates the credentialing and connectivity path but was not yet used to drive the GQE energy loop itself in this submission (see §5).

2.3 GQE Training Results (Table 1)

The Transformer-generated Givens ansatz was trained with $n_{\text{layers}} = 2$, Adam optimizer, learning rate 5×10^{-2}, and central finite-difference gradients ($\varepsilon = 10^{-3}$) chained into the generator via `angles.backward()`, in two separate runs: (i) a 120-step sequential run training H_2 then LiH with one shared model (notebook §7), and (ii) an independent 80-step-per-rung curriculum starting a fresh model and warm-starting up the ladder H_2 →LiH→BeH_2 →H_2O (notebook §8). H_2 (4 qubits, CAS(2,2)/STO-3G) converged to -1.116684 Ha, an error of **20.586 mHa** versus FCI, in both runs – short of the 1.6 mHa chemical-accuracy threshold. **LiH (6 qubits, CAS(2,3)/STO-3G) reached** -7.862027 **Ha, an error of 1.054 mHa, in both runs – inside chemical accuracy**, though in both cases the energy was already at this value at step 0 (inherited from the H_2-trained generator) and did not move further over the following $\sim$80–120 steps, so this reflects the warm-started generator's output rather than LiH-specific training progress. The curriculum then completed BeH_2 (10 qubits, CAS(4,5)/STO-3G) at -15.560312 Ha, an error of **15.389 mHa** – not chemical accuracy, but a genuine completed result, not a failure. It then failed to build the H_2O Hamiltonian entirely (§4), so no H_2O or 4CzIPN energy was obtained. Across all systems that did train, energy stopped changing after approximately step 20, indicating the optimizer reached a stationary point of the finite-difference-estimated gradient rather than the true energy minimum.

System	Qubits	2-qubit gates (n_{layers}=2)	GQE energy error vs. FCI	Chem. acc. (1.6 mHa)?
H_2/STO-3G (exact ansatz, untrained smoke test)	4	4	0.0007 mHa	Yes
H_2/STO-3G (Transformer-trained)	4	4	20.586 mHa	No
LiH/STO-3G (Transformer-trained, warm-started)	6	8	1.054 mHa	Yes
BeH_2/STO-3G (curriculum, warm-started)	10	16	15.389 mHa	No
H_2O/STO-3G	12	20	—	Not reached: Hamiltonian build failed, see §4
4CzIPN CAS(20,20)	40	76	—	Gated, not executed

Table 1: Phase 3 results as actually obtained. 2-qubit gate counts are computed directly from `two_qubit_gate_count(n_qubits, n_layers=2)`. Note the curriculum in this codebase uses STO-3G active spaces (LiH: CAS(2,3)→6 qubits; BeH_2: CAS(4,5)→10 qubits; H_2O: CAS(6,6)→12 qubits), smaller than the cc-pVDZ target sizes quoted in our Phase 2 resource table – this discrepancy is discussed in §4.

2.4 Classical Baseline Comparison

FCI (exact sparse diagonalization via PySCF/OpenFermion) is computed for every system as ground truth and is the baseline against which all errors above are reported; it is exact by construction and thus establishes the correctness ceiling rather than a competing approximate method. The VQE (hardware-efficient Ry+CNOT, matched parameter count) and DMRG (ITensor) baselines specified in the Phase 2 proposal are implemented as described conceptually but were not wired into an executable cell in this snapshot, and are not reported here as run results – we would rather report this gap than present baseline numbers we did not actually generate.

3 Technical Challenges and Mitigation Strategies

Training plateau. The dominant open issue is that GQE training on H_2 and LiH stops improving after $\sim$20 steps, well short of chemical accuracy, even though §2.1 shows the underlying ansatz can reach the exact minimum. Because central finite-difference gradients with $\varepsilon = 10^{-3}$ are used by default, our leading hypothesis is that the gradient estimate becomes too noisy or too small relative to the Adam learning rate near this point, causing the optimizer to stall rather than diverge or oscillate (energy is exactly constant step-to-step, not merely slow-moving). Planned mitigation: switch to the parameter-shift estimator already implemented in `energy_grad(..., method="parameter-shift")`, which is exact for this generator rather than a finite-difference approximation, and sweep the learning rate and finite-difference step size as a diagnostic before attributing the stall to the generator architecture itself.

H_2O Hamiltonian construction failure (diagnosed). The curriculum (§8) successfully completed H_2, LiH, and BeH_2 and failed specifically when building the H_2O Hamiltonian, raising `AssertionError: assert nvir >= 0` inside PySCF's `CASCI.check_sanity()`. We traced this to an infeasible active-space specification: STO-3G water has only 7 total molecular orbitals; with $n_{\text{electron}} = 10$ and the requested active space CAS(6,6) ($\texttt{ncore} = (10-6)/2 = 2$), the number of virtual orbitals is $n_{\text{vir}} = 7-2-6 = -1 < 0$, which is physically infeasible – there are not enough basis functions to support 2 frozen core orbitals plus a 6-orbital active space at this basis size. This also left the benchmark-table cell unable to run (its input `clog` is only assigned if `run_curriculum` returns successfully). Mitigation: either enlarge the basis set for H_2O (e.g., to cc-pVDZ, matching our original Phase 2 resource table) or reduce the requested active space (e.g., CAS(6,5) or CAS(4,6)) so that `ncore + ncas` does not exceed the available orbital count; we will also add a pre-flight feasibility check to `build_hamiltonian` so future misconfigurations raise a clear, actionable error instead of a low-level PySCF assertion.

Active-space size mismatch. The curriculum as coded uses STO-3G active spaces smaller than the cc-pVDZ target sizes in our Phase 2 resource table (Table 1 caption). This does not affect the 40-qubit 4CzIPN target itself, which is defined independently as CAS(20,20), but means intermediate curriculum rungs are currently validating transfer learning at smaller scale than originally planned. Mitigation: align `CURRICULUM` basis/active-space choices with the Phase 2 resource table before the next training pass.

No GPU allocated in this run. All results above used the exact `qpp-cpu` statevector backend, not `tensornet-mps`, because no GPU was available on the qBraid session used for this submission. This caps feasible qubit count well below 40 and means MPS bond-dimension behavior (Table 1 of our Phase 2 proposal, $D = 64$–256) is unvalidated in this snapshot. Mitigation: this is planned to be resolved when the finalist QC-credit allocation is applied to an A100-backed qBraid instance.

4 Quantum vs. Classical: What This Submission Does and Does Not Show

All energy evaluations in this submission were performed on classical simulators (exact statevector via `qpp-cpu`), not on quantum processing hardware, and not yet on `tensornet-mps`. This is consistent with our staged plan: validate ansatz correctness and generator training at small scale on exact simulation before committing GPU/QPU budget to the 40-qubit target. The QCI Dirac-3 connectivity check (§2.2) demonstrates the credentialing and API path needed for a downstream polynomial-optimization role (e.g., post-selection or parameter refinement over the generator's high-sensitivity angles), but Dirac-3 was not yet used to influence any energy reported here. Consequently, this submission does not yet demonstrate quantum advantage or even completed classical-simulator chemical accuracy; its contribution is a validated, reproducible pipeline scaffold plus a precisely characterized failure mode (the training plateau) that the mitigation plan in §4 directly targets.

5 Resource Usage and Reproducibility

Environment. qBraid CPU instance; Python packages `cudaq`, `pyscf`, `openfermion`, `openfermionpyscf`, `torch`, `matplotlib`, `qci-client`$\geq$4 (install cell in notebook §0). **Entry point.** The notebook itself, run top to bottom, is the current reproducibility entry point; the `run_gqe.py --molecule 4CzIPN --qubits 40` CLI mirror described in our Phase 2 proposal has not yet been extracted into a standalone script and is corrected here rather than re-promised. **Wall-clock time** was not instrumented in this run (no per-step timers were logged), so we do not report a specific runtime figure; adding timing instrumentation is a planned fix ahead of the next run. **Disk/memory.** Negligible for the systems executed (4–12 qubits); the 40-qubit target's $\sim$100 MB per energy evaluation at $D = 256$ (Phase 2 Table 1) remains a projection pending GPU access.

6 Conclusion and Next Steps

Phase 3 execution confirms the ansatz and Hamiltonian-construction layers of the proposed MPS-GQE pipeline are physically correct (0.0007 mHa smoke-test error) and that the QCI Dirac-3 integration path is live and authenticated. The transfer-learning curriculum successfully completed H_2, LiH, and BeH_2 – with LiH reaching chemical accuracy (1.054 mHa) – before an infeasible active-space configuration for H_2O, diagnosed precisely in §4, stopped the ladder before it could reach 4CzIPN. Separately, H_2 training plateaus at 20.586 mHa regardless of run, a specific, diagnosable issue distinct from the H_2O configuration bug. Our immediate next steps, in priority order, are: (1) fix the H_2O active-space configuration (§4) and resume the curriculum through H_2O; (2) switch to parameter-shift gradients and re-run H_2 to test the stall hypothesis in §4; (3) wire in the FCI-validated VQE and DMRG baselines already specified in Phase 2; (4) once chemical accuracy is reached at H_2O scale on `tensornet-mps` with GPU access, proceed to the gated 40-qubit 4CzIPN run. We report these as open items rather than completed results because reproducibility and honest limitation reporting are, per the challenge rubric, weighted alongside raw performance.

References

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